

# G. Jagadeesh

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## SUMMARY

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Detail-oriented Computer Science undergraduate with hands-on experience in backend development, microservices architecture, Docker containerization, and cloud deployment. Strong understanding of REST APIs, object-oriented programming, and system design. Actively working with AWS (EC2, DynamoDB), Docker, and Nginx to build scalable and cost-efficient applications. Exploring Generative AI and agentic systems using Python and the Model Context Protocol (MCP).

## SKILLS

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**Languages:** Python, C, JavaScript, SQL

**Frameworks:** Django, Django REST Framework, FastAPI (learning)

**Databases:** PostgreSQL, AWS DynamoDB, SQL

**Cloud & DevOps:** AWS (EC2, DynamoDB, Serverless), Docker, Nginx

**Other:** JWT Auth, Role-Based Access Control, REST APIs, MCP (Model Context Protocol)

## SELF-DESCRIPTION

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**Backend & Cloud Engineer** | Self-Driven Projects | Hyderabad, India | 2022 – Present

- Built and deployed scalable backend services using Django and DRF, containerized with Docker for consistency across environments.
- Optimized AWS Free Tier usage by deploying multiple containers on a single EC2 instance with Nginx as reverse proxy gateway, significantly reducing infrastructure costs through container-based microservices architecture.
- Designed and integrated a Model Context Protocol (MCP) server exposing 31 AI-callable tools across product, cart, orders, and review modules — enabling AI agents to interact with e-commerce APIs in real time (agentic workflow).

## EDUCATION

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**St. Martin's Engineering College**

*Bachelor of Technology — Computer Science*

CGPA: 8.04 / 10

Secunderabad, Hyderabad

*Oct 2022 – May 2026*

## PROJECTS

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### E-Commerce Backend API

- Developed a RESTful backend for an e-commerce platform using Django REST Framework with JWT-based authentication and role-based access control.
- Implemented product management, cart, wishlist, orders, reviews, and Q&A modules with full CRUD support.
- Used PostgreSQL as the primary relational database.
- Containerized the backend using Docker and deployed on AWS EC2.
- Built an MCP (Model Context Protocol) server exposing 31 AI-callable tool endpoints, enabling autonomous AI agent interactions with the platform — a real-world agentic workflow.
- GitHub: <https://github.com/jagadeesh-sagar/ecommerce>

### Microservices Deployment with Docker & Nginx on AWS EC2

- Architected a multi-container microservices deployment on AWS EC2 using Docker, with Nginx as a reverse proxy routing traffic between independent services.
- Each service runs in an isolated container, enabling independent scaling and deployment without shared state.